Biology Class 11

National Book Foundation

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Chapter-wise | SLO Based | Short and Long Questions with Answers

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Chapter 1 contents

- Cells - the basic unit of life, Cell theory
- Microscopy (light and electron microscope)
- Ultrastructure, cell wall, plasma membrane, organelles
- Cell signaling, membrane transport, endo/exocytosis
- Prokaryotic vs eukaryotic cells, Stem cells
- Cell division: mitosis and meiosis
- SLO based Short Questions and Long Questions with Answers

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Chapter 1: Cell and Sub-Cellular Organelles

SLO 1: Cells - the basic unit of life (1.1)

Cell: *The structural and functional unit of all living organisms. It is the smallest unit and building block of life.*

- *Single celled organisms: **Amoeba**, **Paramecium**, **Euglena**. Plants and animals are multicellular.*
- *Shape and size of a cell are related to the function it performs.*
- *A cell shows all **7 properties of life**:*

Property	Meaning
Movement	Living things show movement externally or internally.
Respiration	Use substances from environment to make energy.
Homeostasis	Maintain a stable internal environment.
Growth	Change in shape or size.
Reproduction	Produce offspring, sexually or asexually.
Excretion	Removal of waste products.
Nutrition	Exchange of materials and gases with the environment.

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SLO 2: Cell theory (1.2)

- ***Robert Hooke*** *(1665): saw cork cells (honeycomb-like empty spaces) with a simple microscope and named them "cells".*
- ***Schleiden*** *(German botanist, 1838): cells are the ultimate units of all plant tissue.*
- ***Schwann*** *(German zoologist): animal cells lack a cell wall and are covered by a membrane. Both combined their views = cell theory.*
- ***Virchow*** *(1855): new cells develop by division of pre-existing cells.*

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Three tenets of cell theory: *(1) Cell is the fundamental unit of structure and function in living things. (2) All organisms are made of one or more cells. (3) Cells arise from pre-existing cells by cell division.*

Modern iteration of cell theory

- *Energy moves through cells.*
- *DNA is transferred from cell to cell, carrying genetic information.*
- *All cells share the same primary chemical makeup.*

Validation of cell theory (evidences)

1. *Cells removed from tissues can survive independently for short periods.*
2. *Nothing smaller than a cell has been found that can live independently.*
3. *Redi and Pasteur showed that cells cannot grow in sealed and sterile conditions.*

Exceptions to cell theory

1. ***Viruses** are acellular, have no cell machinery; alive only inside the host cell, behave dead outside.*
2. ***Viroids** and **prions** also behave like viruses.*
3. ***RBCs** and **sieve tube cells** lack nuclei, cannot divide.*
4. ***Bacteria, cyanobacteria** lack a well organized nucleus (no nuclear membrane, nucleolus, nucleoplasm); DNA lies in direct contact with cytoplasm.*
5. ***Coenocytic hyphae** of Rhizopus and the alga Vaucheria are multinucleate.*
6. ***Protozoans** are acellular; body is not divided into cells.*

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SLO 3: Microscopy (1.3)

Microscopy: *The technique used to view objects that cannot be seen by the naked eye.*

Light (compound) microscope

- *Uses visible light and a two-lens system: **ocular lens** (eyepiece) and **objective lens** (revolving nosepiece).*
- *Working: light illuminates the object; the objective forms a real, inverted, magnified first image; this image acts as the object for the eyepiece which*

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forms the final virtual, magnified image (inverted w.r.t. object).

Magnification: *Ability of an optical instrument to increase the size of an object. Total magnification = objective value x ocular value. Max for light microscope = 1500X (100X objective x 15X ocular).*

Resolution: *Ability of a microscope to distinguish between two close points as separate. Naked eye = 0.1 mm; light microscope = 250 nm; electron microscope = 0.2 nm.*

- *Magnification makes small objects look larger; resolution distinguishes two objects from each other. Increasing magnification does not increase resolution.*

Electron microscope

- *Uses a beam of accelerated electrons; very high resolution. Magnifies 1 to 50 million times.*
- *Working: electron gun produces beam, focused by two condenser lenses on a thin specimen section (20-100 nm). Denser regions scatter more electrons. The beam passes through the objective lens to give a magnified image, further magnified by the ocular lens. Accelerating potential is applied between filament and anode.*
- *Two types: **TEM** (transmitted electrons form image) and **SEM** (secondary / backscattered electrons form image). TEM + SEM in one instrument = STEM.*

Light Microscope	**Electron Microscope**
Light (400-700 nm) is the source	Electron beam (approx. 1 nm) is the source
Magnification 1500X	1 to 50 million times
Live and dead specimens can be seen	Only dead / dried specimens
Image by light absorption	Image by electron scattering
Low resolution (250 nm)	High resolution (0.2 nm)
Image is usually coloured	Image is black and white
Vacuum not required	Vacuum is essential
Useful magnification 500X-1500X	Direct magnification up to 16000X, photographic up to 1000000X

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Svedberg unit (S): *Measure of sedimentation rate of a particle when centrifuged; larger value = larger / heavier compact particle. Named after Theodor Svedberg (Nobel Prize 1926).*

SLO 4: Ultrastructure of animal and plant cells (1.4)

Ultrastructure: *The fine structure of the cell as seen with the electron microscope.*

Plant cell	Animal cell
Cell wall present (cellulose)	No cell wall (probably due to locomotor mode of life)
Chloroplasts present in green parts	Chloroplasts absent
Large central vacuole in mature cell	Small or temporary vacuoles
Centrioles generally absent	Centrioles present (mainly in animal cells)
Nucleus pushed away from centre by vacuole	Nucleus generally at the centre

SLO 5: Cell wall and plasma membrane (1.5)

Cell wall

- *Extracellular structure of plant cells, secreted by the cell. Present in plants, prokaryotes, fungi and some protists; absent in animal cells.*
- *Chemical nature: plants = **cellulose**; prokaryotes = **peptidoglycan**; fungi = **chitin**.*
- *Functions: protects the cell, maintains shape, prevents excessive water uptake, supports the plant against gravity. It is porous (permeable) and allows free passage of water and dissolved materials.*
- *Three layers: primary wall, middle lamella, secondary wall.*

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Layer	Details
Primary cell wall	True wall of newly growing cells, thin and flexible, cellulose microfibrils in a matrix of hemicellulose and pectin, crisscross arrangement gives strength, stretches plastically (irreversibly). Adapted for growth.
Secondary cell wall	Between primary wall and plasma membrane, only in sclerenchyma cells (dead, support). Formed after cell reaches maximum size. Cellulose, hemicellulose, lignin, inorganic salts, waxes. Lignin cements microfibrils and gives rigidity.
Middle lamella	Between primary walls of adjacent cells, holds cells together. Sticky gel of calcium and magnesium salts and pectin.

Plasma membrane (cell membrane)

- Boundary of protoplasm; found in all prokaryotic and eukaryotic cells; controls passage of materials into and out of the cell.
- Composition: **proteins 60-80%**, **lipids 20-40%**, small amount of carbohydrates. About 7 nm thick.

Fluid mosaic model: Membrane is a phospholipid bilayer in which protein molecules are partly or wholly embedded and scattered irregularly like icebergs floating in the sea.

- Phospholipids: hydrophobic tails face each other inside, hydrophilic heads on the surfaces. Cholesterol is wedged in the bilayer; it stabilizes the bilayer and restricts entry/exit of polar molecules and ions.
- More unsaturated fatty acids = more fluid, flexible membrane. Fluidity depends on phospholipids, glycolipids and cholesterol.
- Membrane is asymmetrical. Carbohydrates are attached to proteins (glycoproteins) or lipids (glycolipids) on the outer side. Cytoskeleton filaments on inner side support the membrane.

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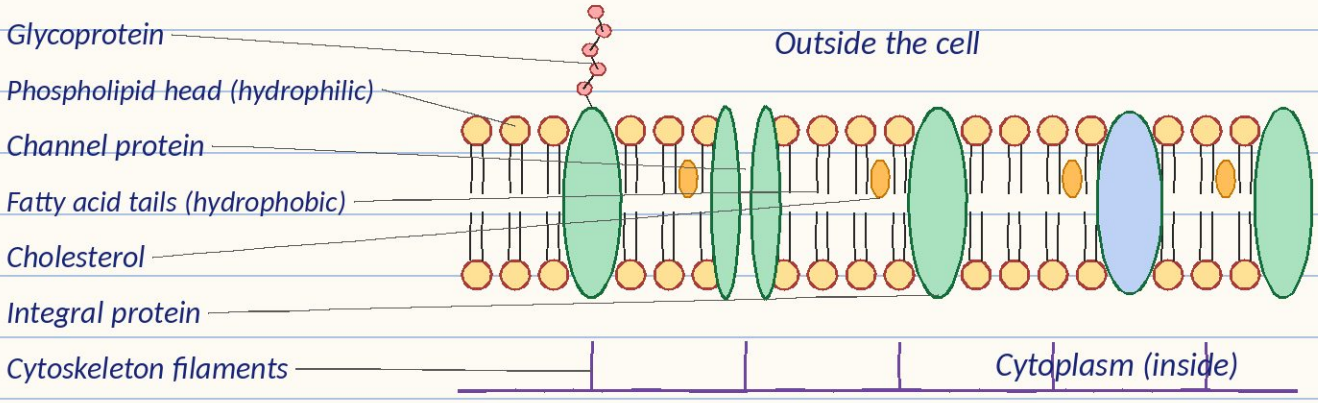

Fig: Fluid mosaic model of plasma membrane

- ***Functions of membrane proteins:*** *channels / carriers for transport, enzymes, receptors (e.g. hormones bind specific receptors), antigens (cell recognition).*
- ***Glycolipids and glycoproteins*** *= cell surface markers; help in cell to cell recognition and holding correct cells together in tissues.*
- ***Regulation of interaction with environment:*** *by controlling transport of materials; maintains suitable pH and ion concentration for enzyme activity.*
- ***Techniques to study membrane:*** *atomic force microscopy, super-resolution fluorescent microscopy, scanning electron microscopy (SEM) which gives morphology/topography and pore size.*

SLO 6: Cell signaling (1.6)

Cell signaling: *The process by which cells communicate with other cells or the external environment; transfer of information from one cell to another.*

- ***Ligands*** *= signaling molecules (proteins, lipids, amino acid metabolites, gases).*
  Receptors *= proteins that respond to ligands; each receptor binds one type of ligand.*

1. ***Reception:*** *target cell detects the signal molecule by binding to a receptor on the surface or inside.*
2. ***Transduction:*** *conversion of the signal into a form that brings a specific response.*
3. ***Response:*** *specific cellular response, e.g. enzyme catalysis, cytoskeleton rearrangement, gene activation.*

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Pathway of signal from outside to inside

- ***Protein signaling:*** *protein/peptide hormones are water soluble, cannot cross the membrane. They act as* ***first messenger****, bind receptors on plasma membrane, generate a* ***second messenger*** *(e.g. cAMP) which activates enzymes, genes etc.*
- ***Steroid signaling:*** *steroids are lipophilic, small, hydrophobic; diffuse freely across the membrane, bind receptors in cytoplasm or nucleus, form hormone-receptor complex, move into nucleus, bind DNA and act as transcription factor. Examples: estrogen, progesterone, testosterone.*

SLO 7: Membrane transport mechanisms (1.7)

- *Four types:* ***simple diffusion****,* ***facilitated diffusion****,* ***osmosis*** *(passive, no energy) and* ***active transport*** *(energy needed).*

Diffusion: *Movement of molecules from higher concentration to lower concentration, down a concentration gradient. Small molecules (O_2, CO_2, water) cross easily. Example: O_2 diffuses from lungs into blood while CO_2 diffuses out.*

Facilitated diffusion: *Diffusion of molecules from higher to lower concentration assisted by a carrier / transport protein. Needs channel proteins (open and gated) or carrier proteins (change shape, affected by temperature and saturation).*

Osmosis: *Movement of water from a solution of low solute concentration to high solute concentration through a semipermeable membrane. Examples: thirst after salty food, swelling of resins and seeds in water.*

- ***Osmotic pressure:*** *minimum pressure applied to a solution to stop flow of solvent through a semipermeable membrane.* ***Water potential:*** *tendency of water to move from one place to another.*

Active transport: *Movement of molecules or ions from lower to higher concentration against the gradient using energy (ATP). Usually in one direction only.*

- ***Primary:*** *energy from ATP directly, e.g.* ***sodium-potassium pump*** *(pumps 3 Na^+*

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out, K^+ in; uses more than a third of cell ATP).

- ***Secondary:** uses electrochemical energy; a transporter couples movement of Na^+ or H^+ down its gradient to move another molecule against its gradient.*
- *Examples: phagocytosis by macrophages, Ca^{2+} out of cardiac cells, amino acids across intestinal lining, secretion of enzymes, hormones, antibodies, plants absorbing minerals by root cells.*

Simple diffusion	***Facilitated diffusion***
Passive transport	*Passive transport*
High to low concentration	*High to low concentration*
No direct ATP needed	*No direct ATP needed*
Transport proteins not required	*Transport proteins required*
Slower but straightforward	*Faster, affected by temperature and protein inhibitors*
Small non-polar molecules (O_2, CO_2)	*Polar molecules (glucose, amino acids), large ions (Na^+, Cl^-)*

SLO 8: Cytoplasm and organelles (1.8)

***Protoplasm:** Living matter of the cell = cytoplasm + nucleus.*

- ***Cytoplasm:** region between nuclear membrane and plasma membrane; about 90% water; contains ions, salts, sugars, amino acids, fatty acids, nucleotides, vitamins, gases and large molecules like proteins (colloidal solution).*
- ***Cytosol** = liquid matrix around organelles. Site of metabolism (protein synthesis, glycolysis) and storehouse of the cell. In prokaryotes all reactions occur in cytosol.*
- ***Cell organelles:** discrete structures in cytoplasm, mostly membrane bound (except ribosomes).*

Endoplasmic reticulum (ER)

- *Network of membranous tubules and sacs (**cisternae**) enclosing the **lumen**. Two types: smooth and rough.*
- ***Smooth ER:** no ribosomes; synthesis of lipids (oil, cholesterol, phospholipids,*

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steroids, sex hormones); detoxifies drugs and poisons in liver. In muscle cells it is called sarcoplasmic reticulum.

- ***Rough ER:** ribosomes on cytoplasmic side; makes secretory proteins which leave in vesicles from **transitional ER**; also a membrane factory (adds proteins and phospholipids to its own membrane).*

Ribosomes

- *Non-membranous granules of rRNA + protein (ribonucleoprotein); site of protein synthesis. Made in nucleolus, exit through nuclear pores.*
- *Eukaryotic = **80S** (60S + 40S); prokaryotic = **70S** (50S + 30S). Subunits held by Mg^{2+} and salt bonds.*
- ***Polysome:** chain of many ribosomes attached to one mRNA; makes many copies of same polypeptide quickly.*

Golgi complex

- *Discovered by Camillo Golgi (1898). Stack of flattened membrane sacs (cisternae) + Golgi vesicles. **Cis face** (forming, convex) receives vesicles from smooth ER; **trans face** (maturing, concave) buds off vesicles.*
- *Functions: processes, modifies and exports cell secretions (proteins); forms glycoproteins, lipoproteins; gives rise to lysosomes, peroxisomes, glyoxysomes; forms vesicles of cell wall material (phragmoplast) in plant cell division.*

Vesicles

- *Small membrane-enclosed fluid sacs (hydrophilic head, hydrophobic tail of phospholipid). Made by Golgi, ER or endocytosis. Types: secretory, transport, synaptic vesicles, lysosomes.*
- *Functions: transport substances within / between cells, metabolism, buoyancy control, enzyme storage, chemical reaction chambers.*

Lysosomes

- *Lyso = splitting, soma = body. Single-membrane vesicles with about 40 hydrolytic (digestive) enzymes, acidic contents. Enzymes made by RER, modified in Golgi, released from trans face as primary lysosomes.*
- *Functions: digestion and removal of waste; **autophagy** (digest own worn*

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organelles; e.g. extra mitochondria after exercise stops); ***autolysis*** *(programmed cell death - lysosomes burst), hence called* ***suicidal bags****. Phagocytosis, endocytosis bring material for digestion.*

Peroxisomes and glyoxysomes (microbodies)

- ***Peroxisomes:*** *single-membrane oxidative vesicles, abundant in liver; enzymes peroxidase, catalase, glycolic acid oxidase. Alcohol detox: alcohol to H_2O_2 (toxic), catalase breaks H_2O_2 into water and oxygen. In plants: photorespiration (glycolate to glycine).*
- ***Glyoxysomes:*** *specialized peroxisomes of oil-seed seedlings; glyoxylate cycle converts stored fats to carbohydrates.*

Vacuoles

- *Large vesicles from ER, Golgi, plasma membrane. Animal: food vacuoles by phagocytosis. Freshwater protists:* ***contractile vacuoles*** *pump out excess water.*
- *Mature plant cell: large* ***central vacuole*** *filled with* ***cell sap*** *(inorganic ions K^+, Cl^-); gives turgor and mechanical support, storehouse. Its membrane =* ***tonoplast****.*

Mitochondria

- *In all eukaryotic cells; number depends on metabolic activity. Self-replicating (binary fission). Cylindrical/rod shaped,* ***double membrane****.*
- ***Outer membrane:*** *smooth, sieve-like with porins.* ***Inner membrane:*** *selectively permeable, folded into* ***cristae*** *(increase surface area) with* ***FO-F1 particles*** *(ATP synthase) and electron transport chain carriers.*
- ***Intermembrane space*** *between membranes.* ***Matrix:*** *jelly-like; small circular DNA, all RNAs, 70S ribosomes, enzymes (own genetic system).*
- *Function: sites of cellular respiration; ATP generation; Krebs cycle in matrix, electron transport chain in inner membrane.*
- *Mitochondria come only from the mother (egg has them, sperm head does not).*

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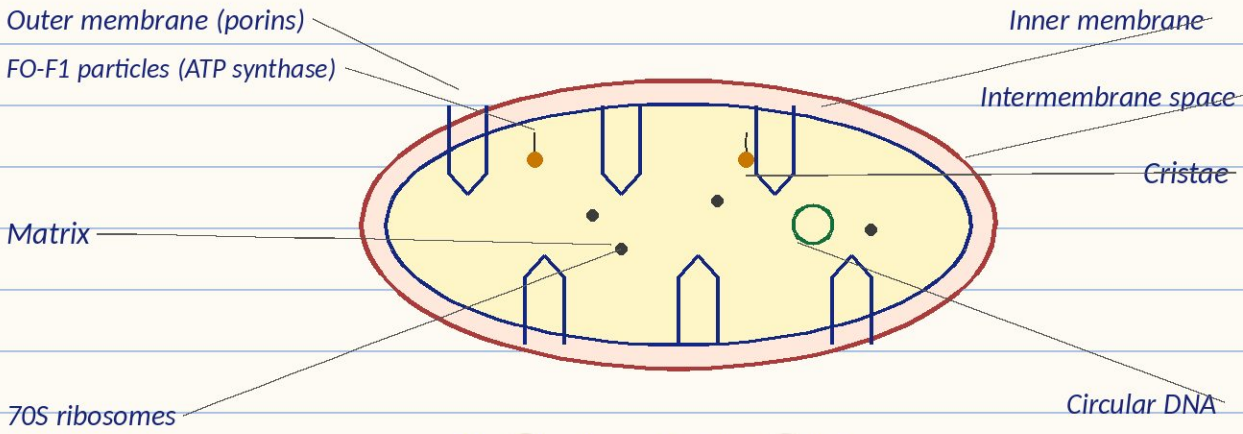

Fig: Structure of mitochondrion

Plastids

- Types: **proplastids** (young, immature), **leucoplasts** (colourless; root, stem, seeds), **chromoplasts** (non-green pigments), **chloroplasts** (green; photosynthesis).
- **Chloroplast:** discoid; envelope (double membrane, outer has porins), **stroma** (colourless ground mass with circular DNA, RNA, 70S ribosomes, enzymes), **thylakoids** (chlorophyll bearing sacs).
- Small thylakoids stacked as **grana** (25-50 thylakoids per granum; 40-60 grana per chloroplast). Large thylakoids (**intergrana**) join grana, colourless.
- Function: **light dependent reaction** in grana (light energy to ATP); **light independent reaction** (dark reaction) in stroma (CO_2 reduced to carbohydrates).

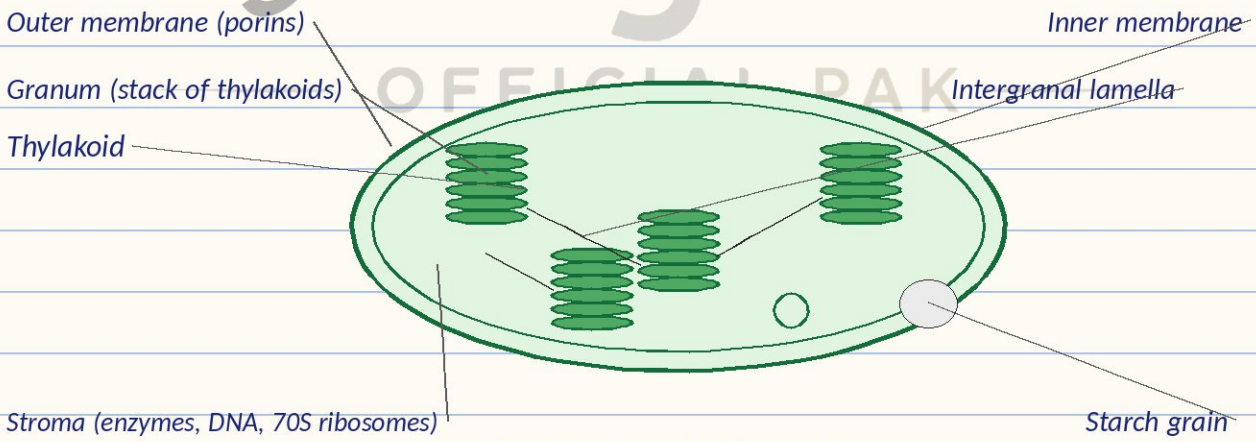

Fig: Structure of chloroplast

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Mitochondria	Chloroplast
Present in all eukaryotic cells	Only in green plant cells
Double membrane; inner membrane folded as cristae	Double membrane; thylakoids, grana in stroma
Matrix with DNA, 70S ribosomes	Stroma with DNA, 70S ribosomes
Site of respiration (ATP made)	Site of photosynthesis (food made)
Have FO-F1 particles	Have chlorophyll pigments

Centrioles and cytoskeleton

- Non-membranous, rod-shaped, in pairs at right angles near nucleus; mainly in animal cells (also slime moulds). Each has **9 triplets of microtubules** in a circle.
- Functions: form spindle fibres (mitotic apparatus) in cell division; give rise to basal bodies of cilia and flagella.
- **Cytoskeleton:** network of microfilaments, microtubules and intermediate filaments.

Nucleus

- Most prominent part of cell: **nuclear envelope, nucleoplasm, nucleolus, chromatin**. (RBCs and sieve tube cells have no nucleus; some cells are multinucleate.)
- **Nuclear envelope:** double membrane with fluid-filled perinuclear space; outer membrane has ribosomes, joined to ER; **nuclear pores** (nucleoporin protein) regulate exchange of RNA, proteins etc.
- **Nucleoplasm:** semifluid ground substance; proteins, enzymes (DNA, RNA polymerase), free nucleotides, Mg^{2+}, histones.
- **Nucleolus:** non-membranous; appears in interphase, disappears in division; granular area (ribosomal subunits) + fibrillar area (rRNA, rDNA); makes ribosomes.
- **Chromatin:** network of DNA + protein threads; condenses into **chromosomes** during division. Chromosome = two **chromatids** joined at **centromere** (with kinetochore proteins).

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SLO 9: Prokaryotic and eukaryotic cells (1.9)

- *Prokaryotes: archaea, bacteria, cyanobacteria. All other organisms are eukaryotic.*

Prokaryotic cell	Eukaryotic cell
No nucleus, no membrane bound organelles	Nucleus and membrane bound organelles present
Always unicellular	Unicellular and multicellular
DNA in cytoplasm	DNA inside the nucleus
One circular chromosome + plasmids	DNA in double stranded chromosomes
Ribosomes 70S (30S + 50S)	Ribosomes 80S (40S + 60S)
Locomotion: flagellin, hook, motor complex	Locomotion: dynein and plasma membrane (cilia, flagella)
Divide by binary fission	Divide by mitosis and meiosis

SLO 10: Stem cells (1.10)

Stem cell: *A cell with the unique ability to develop into specialized cell types of the body. It divides to form daughter cells which become new stem cells or specialized cells (differentiation).*

- **Advantages:** *provide new cells as we grow; replace damaged or lost cells; divide again and again; therapy for spinal cord injury, type 1 diabetes, Parkinson's, Alzheimer's, heart disease, stroke, burns, cancer, osteoarthritis.*

Type	Potential
Adult stem cells	Multipotent: change into only some specific cells. Supply new cells for growth and repair.
Embryonic stem cells (ESCs)	Pluripotent: can produce every cell type of the body. Give new cells for growing embryo.
Induced pluripotent (iPSCs)	Made in laboratory; can develop into any cell type.

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Classification by potential

- ***Totipotent:*** *all cell types (first cells of zygote).* ***Pluripotent:*** *almost any cell (early embryo).* ***Multipotent:*** *closely related family (hematopoietic stem cell gives RBC, WBC, platelets).* ***Oligopotent:*** *a few types (lymphoid / myeloid).* ***Unipotent:*** *only own type, can still self-renew (muscle stem cells).*

iPSCs

- *First described by* ***Shinya Yamanaka*** *(Kyoto University, 2006); Nobel Prize 2012. Made by re-introducing signals that switch off specialization genes and switch on stem cell genes.*
- ***Advantages:*** *give almost any cell type; avoid ethical concerns of human ESCs; cells from the patient give organs with lower risk of rejection.*
- ***Disadvantages:*** *retroviruses used as vectors may insert DNA anywhere and trigger cancer-causing genes; overgrowth of transplanted cells (cancer risk).*

SLO 11: Endocytosis and exocytosis (1.11)

- *Active bulk transport of large molecules, bacteria and wastes that cannot cross the lipid bilayer.*

Endocytosis: *Substances are engulfed by the cell membrane which forms a vesicle containing the material. Two types: phagocytosis and pinocytosis.*

- ***Phagocytosis*** *(cell eating): cell engulfs particles / cells with pseudopodia, forming a phagosome. Amoeba, WBCs. Defensive mechanism in higher animals.*
- ***Pinocytosis*** *(cell drinking): membrane invaginates and buds off a vesicle (pinosome) taking in ions, enzymes, hormones.* ***Receptor mediated endocytosis*** *is a kind of pinocytosis (e.g. cholesterol uptake through coated pits).*

Exocytosis: *Transfer of material from inside to outside of the cell. Vesicles packaged in the cell fuse with the plasma membrane and release contents outside. Examples: neurotransmitter release, sweat gland secretions.*

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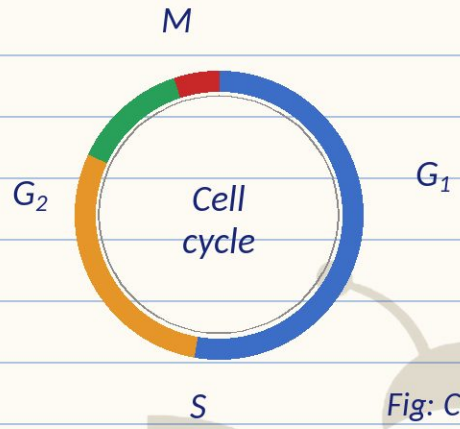
SLO 12: Cell division (1.12)

- *Parent cell divides into two daughter cells. Somatic cells divide by **mitosis**; germ line cells by **meiosis**.*



Interphase = G_1 + S + G_2 (longest)

G_1: first gap, cell grows

S: DNA / chromosomes duplicate

G_2: second gap, prepares for division

M phase = mitosis + cytokinesis (shortest)

Fig: Cell cycle

- ***Mitosis** = division of nucleus; **Cytokinesis** = division of cytoplasm.*

***Mitosis:** Division of the cell in such a manner that chromosomes are duplicated and distributed equally to daughter cells. (Greek mitos = thread.)*

Phase	**Main events**
Interphase	Nuclear envelope present; centrosome replicates (2 centrosomes, each with 2 centrioles); chromosomes duplicated in S phase but not visible.
Prophase	Chromatin condenses into chromosomes (two sister chromatids); nucleoli disappear; mitotic spindle forms; asters; centrosomes move apart.
Prometaphase	Nuclear membrane breaks; spindle invades nucleus; kinetochores form; kinetochore microtubules attach.
Metaphase	Longest stage (about 20 min). Chromosomes line on metaphase plate; centromeres on plate; kinetochores attached to opposite poles.
Anaphase	Shortest stage. Sister chromatids separate (now chromosomes), move centromere first to opposite poles as kinetochore microtubules shorten; cell elongates.
Telophase	Two daughter nuclei form; nuclear envelopes reform; chromosomes decondense.

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Phase	Main events
Cytokinesis	Cytoplasm divides by pinching of membrane (cleavage furrow in animal cell).

- ***Significance of mitosis:*** *growth and multiplication; daughter cells inherit the same chromosome number; continuity of metabolism; wound healing, regeneration, replacement of cells; if uncontrolled may cause tumour / cancer.*

Meiosis: *A single cell divides twice to produce four cells each with half the original genetic information. Gametes (sperms, eggs) in animals; spores in plants.*

Meiosis I (reductional division)

- ***Interphase I:*** *DNA copied; two centrosomes with centriole pairs.*
- ***Prophase I (longest):*** *chromosomes become visible; homologous chromosomes pair lengthwise =* ***synapsis****; each pair =* ***tetrad / bivalent*** *(4 chromatids, 2 centromeres); X-shaped* ***chiasma*** *forms and segments are exchanged =* ***crossing over****; then homologues repel; nucleoli disappear. The* ***diad*** *(two chromatids on one centromere) behaves as one unit.*
- ***Metaphase I:*** *nuclear membrane breaks, spindle forms; homologous pairs line up in double row at equator; each homologue attaches to one pole.*
- ***Anaphase I:*** *homologous chromosomes pulled to opposite poles as spindle fibres shorten; each pole gets one member of each pair.*
- ***Telophase I and cytokinesis:*** *two haploid daughter cells form.*

Meiosis II (like mitosis)

- *Interphase II is brief with no DNA duplication; Prophase II simple; Metaphase II and Anaphase II: centromere divides, sister chromatids separate (become chromosomes) and move to poles; Telophase II: nuclei reform; four haploid cells result.*
- ***Significance of meiosis:*** *keeps chromosome number constant from generation to generation; gives genetic variability; chance for best adapted individual; raw material for evolution.*

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Chapter 1: Short Questions with Answers (SLO based)

Q1. [SLO 1] Define cell. Why is it called the basic unit of life?

Ans: A cell is the structural and functional unit of all living organisms. It is the smallest unit and building block of life, and it carries out all seven properties of life.

Q2. [SLO 1] List the seven properties of life.

Ans: Movement, respiration, homeostasis, growth, reproduction, excretion and nutrition.

Q3. [SLO 1] What is homeostasis? What is nutrition (as given in the chapter)?

Ans: Homeostasis: living things maintain a stable internal environment. Nutrition: living things exchange materials and gases with the environment.

Q4. [SLO 2] State the three tenets of cell theory.

Ans: (1) Cell is the fundamental unit of structure and function in living things. (2) All organisms are composed of one or more cells. (3) Cells arise from pre-existing cells through cell division.

Q5. [SLO 2] Write the contribution of Hooke, Schleiden, Schwann and Virchow.

Ans: Hooke (1665) saw cork cells and named them cells. Schleiden (1838) said cells are the units of plant tissue. Schwann studied animal cells (covered by membrane, no wall). Virchow (1855) said new cells arise from pre-existing cells.

Q6. [SLO 2] How is cell theory validated?

Ans: Cells removed from tissues survive independently for a short time; nothing smaller than a cell can live independently; Redi and Pasteur showed that cells cannot grow in sealed, sterile conditions.

Q7. [SLO 2] Give any four exceptions to cell theory.

Ans: Viruses (acellular), viroids and prions, RBCs and sieve tube cells (no nucleus), bacteria (no organized nucleus), coenocytic hyphae of Rhizopus (multinucleate),

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protozoans (acellular).

Q8. [SLO 2] State the modern iteration of cell theory.

Ans: Energy moves through cells; DNA is transferred from cell to cell carrying genetic information; all cells share the same primary chemical makeup.

Q9. [SLO 3] What is the difference between resolution and magnification?

Ans: Magnification is the ability to make small objects look larger. Resolution is the ability to distinguish two close points as separate. Resolution cannot be increased by increasing magnification. LM resolution = 250 nm, EM = 0.2 nm; LM max magnification 1500X, EM 1 to 50 million times.

Q10. [SLO 3] Why is a light microscope called a compound microscope?

Ans: Because it uses two lenses, the ocular lens in the eyepiece and the objective lens in the nosepiece.

Q11. [SLO 3] Write the working principle of light microscope.

Ans: Objective lens forms a real, inverted, magnified first image. It acts as object for the eyepiece, which forms the final virtual magnified image (inverted w.r.t. the object).

Q12. [SLO 3] Write any four differences between light and electron microscope.

Ans: LM uses light, EM uses electron beams. LM shows live and dead specimens, EM only dead. LM has low resolution, EM high. LM image is coloured, EM black and white. EM needs vacuum, LM does not.

Q13. [SLO 3] What is the difference between TEM and SEM?

Ans: TEM forms the image from transmitted electrons; SEM forms the image by detecting secondary or backscattered electrons.

Q14. [SLO 3] What is the Svedberg unit?

Ans: A measure of sedimentation rate of a particle on centrifugation; indicates its size. Named after Theodor Svedberg.

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Q15. [SLO 4] What is ultrastructure? Name three organelles revealed by the electron microscope.

Ans: The fine structure of the cell seen with the electron microscope. Examples: ribosomes, endoplasmic reticulum, Golgi complex, mitochondria.

Q16. [SLO 5] Why is the cell wall not present in animal cells?

Ans: It is probably due to their locomotor mode of life. Animal cells are covered only by a plasma membrane.

Q17. [SLO 5] Compare the chemical nature of the cell wall in plants, prokaryotes and fungi.

Ans: Plants: cellulose. Prokaryotes: peptidoglycan. Fungi: chitin.

Q18. [SLO 5] Name the three layers of the plant cell wall.

Ans: Primary cell wall, middle lamella and secondary cell wall.

Q19. [SLO 5] What is the role of lignin?

Ans: Lignin cements and anchors cellulose microfibrils together in the secondary wall and makes it rigid.

Q20. [SLO 5] Is the plant cell wall permeable or impermeable?

Ans: It is porous and permeable; it allows free passage of water and dissolved materials.

Q21. [SLO 5] Write the chemical composition of plasma membrane.

Ans: Proteins 60-80%, lipids 20-40% and a small quantity of carbohydrates.

Q22. [SLO 5] What is the fluid mosaic model?

Ans: The membrane is a phospholipid bilayer in which protein molecules are partly or wholly embedded and scattered like icebergs floating in the sea.

Q23. [SLO 5] What is the role of cholesterol in the membrane?

Ans: It is wedged in the bilayer, stabilizes it, and restricts entry and exit of polar molecules and ions.

Q24. [SLO 5] What is the role of glycolipids and glycoproteins as cell surface markers?

Ans: They are involved in cell to cell recognition and in sticking the correct cells together in tissues.

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Q25. [SLO 5] What techniques are used to study the plasma membrane?

Ans: Atomic force microscopy, super-resolution fluorescent microscopy and scanning electron microscopy (SEM).

Q26. [SLO 6] Define cell signaling, ligand and receptor.

Ans: Cell signaling: transfer of information from one cell to another. Ligand: signaling molecule that initiates signaling. Receptor: protein that responds to a ligand.

Q27. [SLO 6] Write the three steps of cell signaling.

Ans: Reception, transduction and response.

Q28. [SLO 6] Why can steroid hormones enter the cell directly?

Ans: They are small, hydrophobic (lipophilic) molecules, so they diffuse freely through the plasma membrane. Examples: estrogen, progesterone, testosterone.

Q29. [SLO 6] What are first and second messengers?

Ans: The protein hormone is the first messenger (binds receptor on the membrane). It generates a second messenger such as cAMP inside the cell.

Q30. [SLO 7] Define diffusion and facilitated diffusion.

Ans: Diffusion: movement of molecules from higher to lower concentration down a gradient. Facilitated diffusion: same movement assisted by channel or carrier proteins.

Q31. [SLO 7] Define osmosis. Give two examples.

Ans: Passage of water from a solution of low solute concentration to high concentration through a semipermeable membrane. Examples: thirst after salty food; swelling of resins and seeds.

Q32. [SLO 7] Define osmotic pressure and water potential.

Ans: Osmotic pressure: minimum pressure applied to a solution to stop solvent flow through a semipermeable membrane. Water potential: measure of the tendency of water to move from one place to another.

Q33. [SLO 7] Define active transport. Give two examples.

Ans: Movement of molecules or ions from lower to higher concentration against the gradient using energy (ATP). Examples: sodium-potassium pump; absorption of

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amino acids in the intestine.

Q34. [SLO 7] What is the sodium-potassium pump?

Ans: A primary active transport pump in animal cells that pumps sodium ions out and potassium ions into the cell using ATP; more than a third of cell ATP is used for it.

Q35. [SLO 7] Differentiate between primary and secondary active transport.

Ans: Primary uses ATP directly. Secondary uses electrochemical energy: a transporter couples movement of Na^+ or H^+ down its gradient with movement of another molecule against its gradient.

Q36. [SLO 8] What is cytosol? What is the chemical nature of cytoplasm?

Ans: Cytosol is the liquid matrix of cytoplasm around organelles. Cytoplasm is about 90% water with ions, small molecules in true solution and large molecules (proteins) forming colloidal solutions.

Q37. [SLO 8] What are the metabolic roles of cytoplasm?

Ans: It is the site of metabolism (protein synthesis, glycolysis) and the storehouse of the cell.

Q38. [SLO 8] Differentiate between smooth ER and rough ER.

Ans: Smooth ER has no ribosomes and makes lipids and steroids, detoxifies drugs. Rough ER has ribosomes on its surface and makes secretory proteins and membrane.

Q39. [SLO 8] What holds the ribosomes together in a polysome?

Ans: A single mRNA molecule. Several ribosomes attach to it and make many copies of the same polypeptide quickly.

Q40. [SLO 8] Why are ribosomes called ribonucleoprotein? Give their S values.

Ans: They are made of almost equal amounts of RNA and protein. Eukaryotic 80S (60S + 40S); prokaryotic 70S (50S + 30S).

Q41. [SLO 8] Explain that nucleoli are the areas where ribosomes are assembled.

Ans: Nucleolus has a granular area with ribosomal subunits and a fibrillar area with rRNA and rDNA. Ribosomes are formed here and move to the cytoplasm through

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nuclear pores.

Q42. [SLO 8] What are vesicles and what is their function?

Ans: Small membrane-enclosed sacs made in Golgi, ER or by endocytosis. They transport substances within and between cells, and help in metabolism, buoyancy control and enzyme storage.

Q43. [SLO 8] Why are lysosomes called suicidal bags?

Ans: In autolysis (programmed cell death) lysosomes burst and release enzymes which digest the cell itself.

Q44. [SLO 8] What would happen if there were no lysosomes in human cells?

Ans: Cells could not digest engulfed material or remove waste and worn organelles (no autophagy), and programmed cell death by autolysis would not occur.

Q45. [SLO 8] Define autophagy.

Ans: The self-eating process in which a lysosome digests the cell's own organelles, e.g. extra mitochondria after heavy exercise stops.

Q46. [SLO 8] What are peroxisomes? State their function in the liver.

Ans: Single-membrane oxidative vesicles containing peroxidase and catalase. In the liver they detoxify alcohol: alcohol is oxidized to H_2O_2, then catalase splits H_2O_2 into water and oxygen.

Q47. [SLO 8] What are glyoxysomes?

Ans: Specialized peroxisomes in seedlings of oil seed plants; they convert stored fats to carbohydrates through the glyoxylate cycle.

Q48. [SLO 8] What are tonoplast and cell sap?

Ans: Tonoplast is the membrane separating the central vacuole from cytoplasm. Cell sap is the solution inside the vacuole, rich in ions such as K^+ and Cl^-.

Q49. [SLO 8] What are cristae and FO-F1 particles?

Ans: Cristae are folds of the inner mitochondrial membrane that increase surface area. FO-F1 particles on them are ATP synthase enzymes.

Q50. [SLO 8] How is a chloroplast similar to a bacterium?

Ans: Both have a small circular DNA, 70S ribosomes and can self-replicate (division like an independent organism).

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Q51. [SLO 8] Name the types of plastids.

Ans: Proplastids, leucoplasts (colourless), chromoplasts (non-green pigments) and chloroplasts (green).

Q52. [SLO 8] What are centrioles? Write their structure.

Ans: Non-membranous rod-shaped organelles in pairs at right angles near the nucleus; each has nine triplets of microtubules arranged in a circle.

Q53. [SLO 8] Write the functions of centrioles.

Ans: They form spindle fibres during cell division and give rise to basal bodies of cilia and flagella.

Q54. [SLO 8] Differentiate between chromatin and chromosome.

Ans: Chromatin is a network of thin DNA-protein threads in the nondividing nucleus. Chromosome is the thick condensed structure of two chromatids joined at a centromere, seen during division.

Q55. [SLO 8] Name organelles that are single membrane, double membrane and non-membranous.

Ans: Single: ER, Golgi, lysosomes, peroxisomes, vacuoles. Double: nucleus, mitochondria, chloroplast. None: ribosomes, centrioles, nucleolus.

Q56. [SLO 8] Compare nucleoplasm with cytoplasm.

Ans: Nucleoplasm is a semifluid with proteins, enzymes (DNA and RNA polymerase), free nucleotides, Mg^{2+}, histones for DNA/RNA synthesis. Cytoplasm is mostly water with metabolic molecules. Nucleoplasm is slightly different from cytoplasm.

Q57. [SLO 9] What are prokaryotic cells? List the structures missing in them.

Ans: Cells of archaea, bacteria and cyanobacteria. They lack a true nucleus and membrane bound organelles (mitochondria, ER, Golgi, lysosomes, plastids).

Q58. [SLO 9] Name the structures common to plant, animal and prokaryotic cells.

Ans: Plasma membrane, cytoplasm, ribosomes and DNA (genetic material).

Q59. [SLO 10] Define stem cell. Write two advantages of stem cells.

Ans: A cell with the unique ability to develop into specialized cell types. Advantages: replace damaged or lost cells; divide again and again; treat diseases like diabetes,

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Parkinson's, spinal cord injury.

Q60. [SLO 10] What are pluripotent stem cells?

Ans: Stem cells that can naturally produce almost every cell type of the body, e.g. embryonic stem cells.

Q61. [SLO 10] Who discovered iPSCs? Why are they useful?

Ans: Shinya Yamanaka (Kyoto University, 2006; Nobel Prize 2012). They give any cell type, avoid ethical issues of ESCs and have a lower risk of rejection.

Q62. [SLO 11] Define endocytosis, phagocytosis, pinocytosis and exocytosis.

Ans: Endocytosis: engulfing substances in a membrane vesicle. Phagocytosis: cell eating by pseudopodia. Pinocytosis: cell drinking, taking fluid droplets by invagination. Exocytosis: release of vesicle contents outside by fusion with the membrane.

Q63. [SLO 12] Name the phases of the cell cycle. When are chromosomes duplicated?

Ans: Interphase (G_1, S, G_2) and M phase (mitosis + cytokinesis). Chromosomes duplicate only in S phase.

Q64. [SLO 12] Define mitosis and cytokinesis.

Ans: Mitosis: division of the cell so that chromosomes are duplicated and distributed equally. Cytokinesis: division of cytoplasm, by cleavage furrow in animal cells.

Q65. [SLO 12] What is the significance of mitosis?

Ans: Growth, same chromosome number in daughter cells, continuity of metabolism, wound healing, regeneration and replacement of cells.

Q66. [SLO 12] Define synapsis, tetrad, chiasma and crossing over.

Ans: Synapsis: lengthwise pairing of homologous chromosomes. Tetrad: a synapsed pair of four chromatids. Chiasma: X-shaped crossing point. Crossing over: exchange of segments between homologues at the chiasma.

Q67. [SLO 12] Why is meiosis important?

Ans: It keeps chromosome number constant, gives genetic variability, and provides raw material for evolution.

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Chapter 1: Long Questions with Answers (SLO based)

Q1. [SLO 5] Describe the location, chemical composition and significance of primary cell wall, middle lamella and secondary cell wall.

Ans:

- ***Primary cell wall:*** *true wall, inner to middle lamella, in newly growing cells. Cellulose microfibrils in a matrix of hemicellulose and pectin; crisscross arrangement gives great strength. Thin and flexible; stretches plastically so it is adapted for growth.*
- ***Middle lamella:*** *between primary walls of adjacent cells. Sticky gel of Ca and Mg salts (pectates) and pectin. Holds the cells together.*
- ***Secondary cell wall:*** *between primary wall and plasma membrane, only in sclerenchyma cells, formed after the cell has stopped growing. Cellulose, hemicelluloses, lignin, inorganic salts and waxes. Lignin makes it rigid. Gives definite shape and mechanical support; the cells are dead.*
- *General functions: protects cell, maintains shape, prevents excess water uptake and holds the plant up against gravity.*

Q2. [SLO 5] Explain (a) chemical composition of plasma membrane (b) role of plasma membrane in regulating the cell's interaction with environment.

Ans:

(a) Composition

- *Proteins 60-80%, lipids 20-40% (phospholipids, glycolipids, cholesterol) and a small quantity of carbohydrates. Carbohydrates are attached to proteins (glycoproteins) or lipids (glycolipids) on the outer surface.*

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(b) Role in regulation

- *It controls transport of material into and out of the cell (diffusion, osmosis, active transport, endo/exocytosis).*
- *It maintains suitable pH and ionic concentration inside the cell for enzyme activity.*
- *Receptor proteins receive signals (e.g. hormones) from the environment and other cells.*
- *Glycoproteins / glycolipids work as markers for cell recognition.*

Q3. [SLO 5] Describe the lipid composition and variety of proteins of the plasma membrane. Write the functions of membrane proteins.

Ans:

- ***Lipids:*** *two layers of phospholipids with hydrophobic tails facing inward and hydrophilic heads on the surfaces; cholesterol wedged between them. More unsaturated fatty acids make the membrane more fluid; cholesterol stabilizes it.*
- ***Proteins:*** *scattered in the bilayer, wholly or partly embedded; pattern differs from membrane to membrane and on both sides. They float sideways in the fluid bilayer.*
- ***Functions:*** *(1) channels and carriers for passage of molecules; (2) enzymes; (3) receptors for hormones and signals; (4) antigens for cell recognition.*

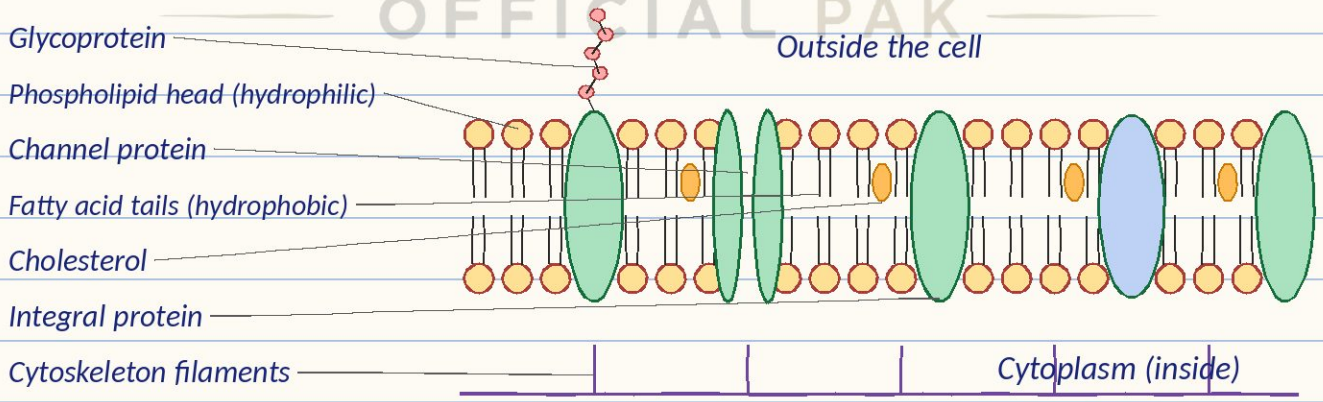

Fig: Fluid mosaic model of plasma membrane

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Q4. [SLO 8] Describe the structures and functions of smooth and rough endoplasmic reticulum. How is ER related to Golgi, lysosomes and plasma membrane?

Ans:

- *ER is a network of membranous tubules and sacs (cisternae) with a lumen separated from cytosol.*
- ***Smooth ER:** continuous with rough ER, no ribosomes. Makes lipids (oil, cholesterol, phospholipids, steroids, sex and adrenal hormones); detoxifies drugs and poisons in liver.*
- ***Rough ER:** ribosomes on the cytoplasmic side. Makes secretory proteins which leave in vesicles budding from transitional ER; adds proteins and phospholipids to its own membrane (membrane factory).*
- ***Relationship:** proteins from RER go via SER and transport vesicles to the Golgi (cis face); Golgi modifies and sends them out as vesicles; lysosomal enzymes are made by RER and sent to Golgi; ER membrane expands and is passed as vesicles to other endomembrane parts including the plasma membrane.*

Q5. [SLO 8] Explain the structure, chemical composition and function of ribosomes.

Ans:

- *Roughly spherical granular non-membranous particles in prokaryotes and eukaryotes; seen only under electron microscope.*
- *Composition: almost equal rRNA and protein, so ribonucleoprotein. Formed in nucleolus and exported through nuclear pores.*
- *Structure: two subunits. Eukaryotes 80S (60S + 40S); prokaryotes 70S. Subunits join with Mg^{2+} and salt bonds between RNA phosphate and amino group of amino acids.*
- *Function: protein synthesis. Many ribosomes on one mRNA form a polysome which makes many copies of a polypeptide in a short time.*

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Q6. [SLO 8] Explain the structure and functions of Golgi complex.

Ans:

- *Found in all eukaryotic cells; discovered by Camillo Golgi (1898); also called Golgi apparatus or Golgi bodies.*
- *Structure: stack of flattened membrane bound sacs (cisternae) with Golgi vesicles.* ***Cis (forming) face*** *convex, receives vesicles from smooth ER;* ***trans (maturing) face*** *concave, cisternae break into vesicles.*
- *Functions: collects proteins from RER via SER, modifies them and exports as vesicles (secretion); makes glycoproteins and lipoproteins; gives rise to lysosomes, peroxisomes and glyoxysomes; in plant cell division its vesicles form the phragmoplast and later the new cell wall. Abundant in secretory cells.*

Q7. [SLO 8] Explain the structure and functions of peroxisomes and glyoxysomes in animal and plant cells.

Ans:

- *Both are single-membrane vesicular microbodies originating from Golgi, smaller than lysosomes; import proteins from cytosol.*
- ***Peroxisomes:*** *contain peroxidases, catalase and glycolic acid oxidase. Abundant in liver; detoxify alcohol: alcohol to H_2O_2, catalase converts H_2O_2 to water and O_2. In plants, a step of photorespiration (glycolate to glycine, by glycolic acid oxidase) occurs here.*
- ***Glyoxysomes:*** *specialized peroxisomes in oil seedlings; enzymes of glyoxylate cycle convert stored fatty acids into carbohydrates in germinating seeds.*

Q8. [SLO 8] Explain the formation, structure and functions of lysosomes.

Ans:

- ***Formation:*** *enzymes are made by RER, passed through SER to Golgi, modified, then released from trans face as vesicles = primary lysosomes.*
- ***Structure:*** *single membranous spherical vesicles with about 40 hydrolytic enzymes; acidic contents; can digest every major macromolecule.*

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- ***Functions:*** *digestion of material from endocytosis / phagocytosis; autophagy (digestion of own organelles in starvation or to control organelle number); autolysis (programmed cell death), so called suicidal bags; removal of waste.*
- *In plants and fungi certain vacuoles perform the same hydrolysis.*

Q9. [SLO 8] Describe the external and internal structure of mitochondrion. What are the functions of these structures?

Ans:

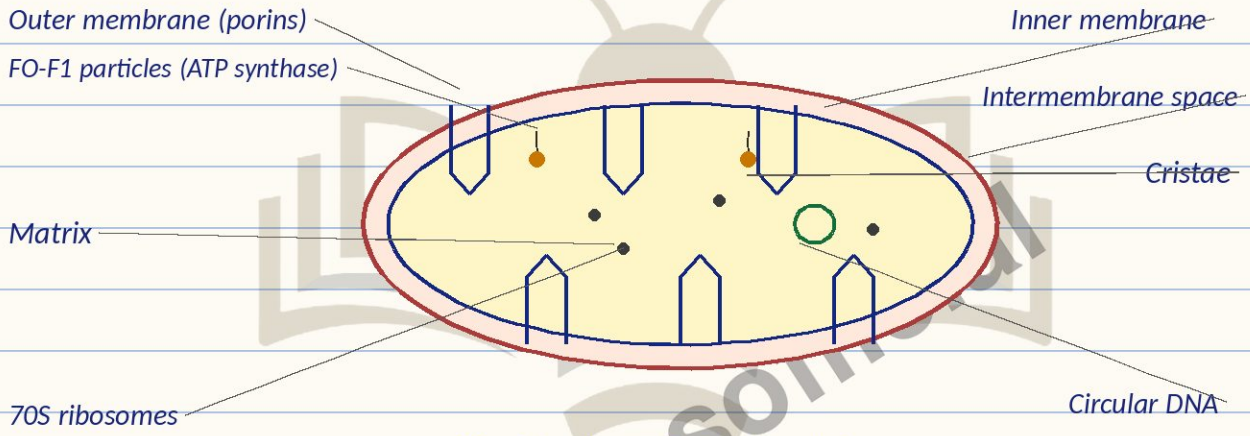

Fig: Structure of mitochondrion

Part	***Structure / Function***
Outer membrane	*Smooth, sieve-like because of porins; lets molecules pass into intermembrane space.*
Inner membrane	*Selectively permeable, folded into cristae; cristae increase surface area; carry FO-F1 particles (ATP synthase) and electron transport chain carriers.*
Intermembrane space	*Narrow region between two membranes.*
Matrix	*Jelly-like; circular DNA, RNAs, 70S ribosomes, enzymes for Krebs cycle; own genetic system.*

- *Function: site of cellular respiration; makes ATP from sugars, fats and other organic compounds using oxygen. Self-replicates by fission.*

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Q10. [SLO 8] Describe the external and internal structure of chloroplast. What are the functions of these structures?

Ans:

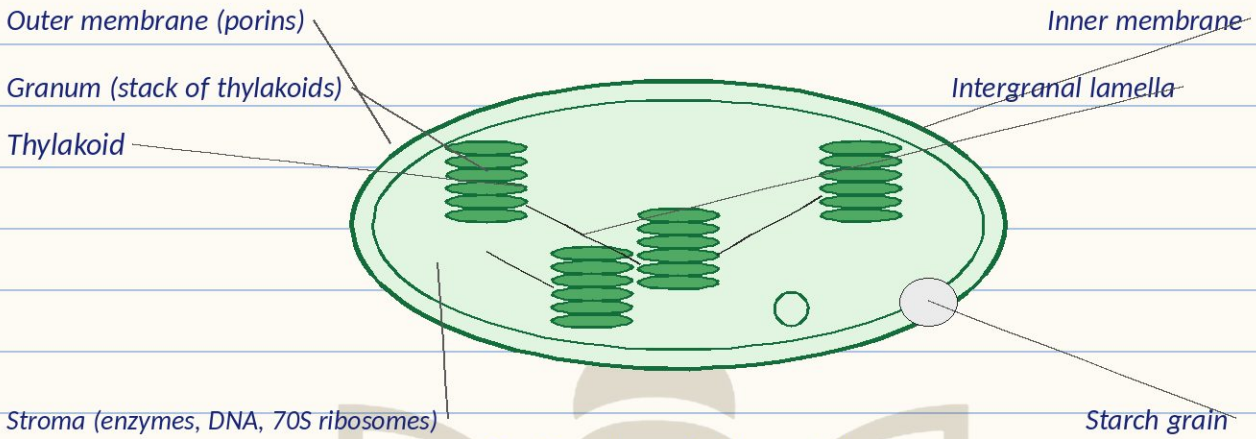

Fig: Structure of chloroplast

Part	Structure / Function
Envelope	Smooth double membrane; outer has porins (freely permeable), inner semipermeable and rich in protein; intermembrane space between.
Stroma	Colourless proteinaceous ground mass; circular DNA, RNAs, 70S ribosomes, enzymes; site of light independent (dark) reaction where CO_2 is reduced to carbohydrates.
Thylakoids	Chlorophyll bearing flattened sacs; small thylakoids stacked as grana (25-50 per granum, 40-60 grana); large thylakoids = intergrana join them.
Grana	Site of light dependent reaction: sunlight captured and changed to ATP.

Q11. [SLO 8] Compare and contrast the structure and functions of mitochondria and chloroplasts.

Ans:

Mitochondria	Chloroplasts
In all eukaryotic cells	In green plant cells and algae
Double membrane; inner folded into cristae	Double membrane; thylakoids stacked in grana
Matrix has DNA, 70S ribosomes, enzymes	Stroma has DNA, 70S ribosomes, enzymes

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Mitochondria	Chloroplasts
Cellular respiration, ATP made	Photosynthesis, food made
FO-F1 particles on cristae	Chlorophyll in thylakoid membranes
Both: have porins in outer membrane, self-replicating, own genetic system	Both: have porins in outer membrane, self-replicating, own genetic system

Q12. [SLO 8] What are centrioles? Describe the structure, composition and functions of centriole.

Ans:

- Non-membranous, rod-shaped organelles mostly in animal cells (and slime moulds, water moulds); usually in pairs at right angles to each other near one pole of the nucleus.
- Each centriole is composed of nine triplets of microtubules arranged in a circle around a central axis.
- Functions: before division the pair duplicates and the pairs move to opposite sides; they give rise to spindle fibres (mitotic apparatus) which distribute chromosomes; they form basal bodies of cilia and flagella.

Q13. [SLO 8] What is nuclear envelope? Describe its chemical composition and structure.

Ans:

- Double membrane covering that forms the boundary of the nucleus; the two membranes are separated by fluid-filled perinuclear space.
- Composition: lipid bilayer and proteins.
- Outer membrane is covered with ribosomes and continuous with ER membranes.
- Nuclear pores made of the transport protein nucleoporin; at pores both membranes are joined. They regulate nucleo-cytoplasmic exchange: RNA and ribosomal proteins out; DNA polymerase, carbohydrates, signals, lipids in. Small molecules diffuse, large ones need signal sequences.

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Q14. [SLO 2] What is cell theory? How is it validated? What are its exceptions?

Ans:

- *Cell theory (Schleiden, Schwann; Virchow added division): (1) cell is the unit of structure and function; (2) all organisms consist of one or more cells; (3) cells arise from pre-existing cells.*
- ***Validation:** cells removed from tissues survive for a short time; nothing smaller than a cell lives independently; Redi and Pasteur: cells do not grow in sealed sterile conditions.*
- ***Exceptions:** viruses, viroids, prions; RBCs and sieve tube cells (no nucleus); bacteria and cyanobacteria (no organized nucleus); multinucleate coenocytic hyphae of Rhizopus and Vaucheria; protozoans (acellular).*

Q15. [SLO 3] Compare and contrast the working of light microscope and electron microscope.

Ans:

- ***LM:** light illuminates specimen; objective forms real inverted magnified image; eyepiece forms final virtual image. Max 1500X, resolution 250 nm, live or dead specimens.*
- ***EM:** electron gun beam, two condenser lenses, thin specimen (20-100 nm), denser regions scatter more electrons, objective and ocular lenses magnify; 1 to 50 million X, resolution 0.2 nm, only dead specimens, vacuum needed.*

Light microscope	***Electron microscope***
Light source (400-700 nm)	*Electron beam (1 nm)*
Low resolution	*High resolution*
Live and dead specimens	*Dead / dried specimens only*
Coloured image, no vacuum	*Black and white image, vacuum needed*

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Q16. [SLO 6] Describe the pathway of protein signal and steroid signal from outside of the cell to inside.

Ans:

Protein (peptide) signal

- *Protein / peptide hormones are water soluble and cannot cross the plasma membrane. The hormone (first messenger) binds its specific receptor on the surface of the target cell.*
- *This starts events inside the cell that make a second messenger, e.g. cAMP. The second messenger activates enzymes and genes = response.*

Steroid signal

- *Steroids are lipids with a four-ring hydrocarbon skeleton; lipophilic so they diffuse across the membrane.*
- *They bind receptors in cytoplasm or nucleus, forming an active hormone-receptor complex, which enters the nucleus and binds DNA as a transcription factor to activate gene expression (estrogen, progesterone, testosterone).*
- *Steps common to both: reception, transduction, response.*

Q17. [SLO 10] Categorize and explain different types of stem cells. What are the advantages and disadvantages of iPSCs?

Ans:

- ***By source:** adult stem cells (multipotent, replace damaged cells), embryonic stem cells (pluripotent, make every cell type) and induced pluripotent stem cells (made in the lab).*
- ***By potency:** totipotent (all cells; first cells of zygote), pluripotent (almost any cell; early embryo), multipotent (related family, e.g. blood cells), oligopotent (few types; lymphoid / myeloid), unipotent (only one type; muscle stem cells).*
- ***iPSC advantages:** can become any cell type; no ethical problems of embryos; patient's own cells reduce rejection risk of new organs.*
- ***iPSC disadvantages:** retroviral vectors may insert DNA anywhere and trigger*

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cancer genes; overgrowth of transplanted cells (cancer risk).

Q18. [SLO 7] Explain simple diffusion, facilitated diffusion, osmosis and active transport. Compare simple and facilitated diffusion.

Ans:

- ***Simple diffusion:*** *small non-polar molecules (O_2, CO_2, water) move directly through the bilayer from high to low concentration; no energy, no proteins.*
- ***Facilitated diffusion:*** *polar molecules and ions move down the gradient through channel proteins (open or gated) or carrier proteins (change shape); no ATP.*
- ***Osmosis:*** *water moves through a semipermeable membrane from low to high solute concentration. Cell in hypertonic solution loses water; in hypotonic gains water.*
- ***Active transport:*** *against the gradient using ATP (primary: Na-K pump) or electrochemical energy (secondary); carrier proteins used; one direction.*

Simple diffusion	Facilitated diffusion
No transport proteins	Transport proteins needed
Slower, depends on gradient	Faster, affected by temperature and inhibitors
Small non-polar molecules	Polar molecules, large ions

Q19. [SLO 9] Differentiate between prokaryotic and eukaryotic cells with diagram.

Ans:

Prokaryotic	Eukaryotic
No nucleus or membrane bound organelles	Nucleus and organelles present
Always unicellular	Uni- and multicellular
DNA in cytoplasm, one circular chromosome + plasmids	DNA in nucleus, double stranded chromosomes
70S ribosomes (30S + 50S)	80S ribosomes (40S + 60S)

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Prokaryotic	Eukaryotic
Flagellin, hook and motor complex	Dynein and plasma membrane
Binary fission	Mitosis and meiosis

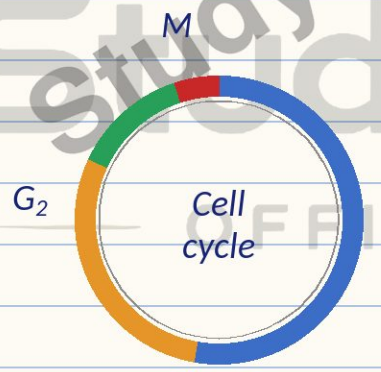
Q20. [SLO 11] Explain endocytosis and exocytosis with diagram.

Ans:

- **Endocytosis:** cell membrane folds in and engulfs substances into a vesicle. (a) **Phagocytosis:** pseudopodia surround a large particle and form a phagosome (Amoeba, WBCs, defence). (b) **Pinocytosis:** cell drinking; membrane invaginates and forms a pinosome; includes receptor mediated endocytosis through coated pits (cholesterol uptake).
- **Exocytosis:** secretory vesicles from Golgi move to the plasma membrane, fuse with it and release their contents outside (neurotransmitters at synapse, sweat gland secretions, enzymes, hormones).
- Both are active bulk transport needing ATP and vesicles.

Q21. [SLO 12] Explain the stages of mitosis with diagram.

Ans:



Interphase = G_1 + S + G_2 (longest)

G_1: first gap, cell grows

S: DNA / chromosomes duplicate

G_2: second gap, prepares for division

M phase = mitosis + cytokinesis (shortest)

Fig: Cell cycle

- **Interphase:** duplicated chromosomes not visible; centrosome duplicates.
- **Prophase:** chromosomes visible (two sister chromatids), nucleoli vanish, spindle and asters form, centrosomes move to poles.
- **Prometaphase:** nuclear envelope breaks, kinetochore microtubules attach to kinetochores.

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- ***Metaphase:*** *chromosomes line on metaphase plate (longest stage).*
- ***Anaphase:*** *centromeres divide, chromatids move to opposite poles (shortest stage).*
- ***Telophase:*** *two nuclei re-form, chromosomes uncoil; then* ***cytokinesis*** *by cleavage furrow.*

Q22. [SLO 12] Explain the stages of meiosis with diagram.

Ans:

- ***Meiosis I:*** *Interphase I (DNA copied). Prophase I (synapsis, tetrad, chiasma, crossing over). Metaphase I (homologous pairs on equator in double row). Anaphase I (homologues separate to poles, number halves). Telophase I + cytokinesis: two haploid cells.*
- ***Meiosis II:*** *like mitosis; no DNA duplication. Metaphase II and Anaphase II: centromeres divide and sister chromatids separate. Telophase II: four haploid cells.*
- ***Result:*** *four haploid cells with genetic variation (crossing over + random distribution).*

Mitosis	***Meiosis***
One division	*Two divisions*
Two daughter cells, same chromosome number	*Four haploid cells*
Somatic cells; growth and repair	*Germ line cells; gametes / spores*
No crossing over	*Crossing over in prophase I*

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